ANTI-MONEY LAUNDERING · 2026 EDITION

# An AML program every decision can be replayed.

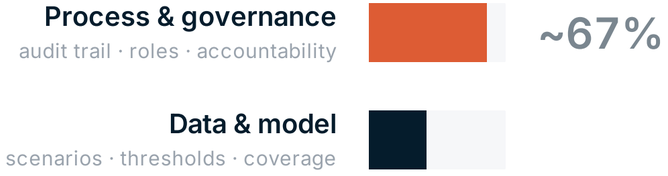
Process and governance gaps outnumber data and model gaps roughly 2:1 in every recent consent order. The cost of being unable to evidence a decision is now larger than the cost of missing one.

PREPARED FOR **CCO · MLRO · AUDIT COMMITTEE**

**Source:** FCA Dear-CEO Letter (Mar 2024) · FinCEN BSA consent order (Oct 2024 · $3.09B) · LexisNexis True Cost of Compliance (Feb 2024)

# Process and governance gaps account for ~2:1 of every consent-order finding — not detection failures.

Five of the largest 2024–25 enforcement actions surveyed. Regulators rarely allege the bank missed a typology. They allege the bank cannot evidence what it did.




> *"Decisions made in relation to financial crime were not supported by evidence or an audit trail of debate and challenge."*
>
> **FCA DEAR-CEO LETTER TO ANNEX 1 FIRMS** · MARCH 2024 · STILL OPERATIVE IN 2026

**KEY TAKEAWAY** Lead with proof, not detection. The single highest-leverage place to spend a compliance dollar in 2026 is on *provability*.

**Source:** Synthesis of the five largest 2024–25 BSA / AML consent orders · FCA Dear-CEO Letter Mar 2024 · K&L Gates 2024 BSA enforcement-action review

# The bill arrives in three sizes — and the largest is still growing.

A single record-breaking penalty in 2024, an industry-wide annual run-rate, and a daily-process tax — the same dollars that buy detection also buy poor evidence and poor signal.

SINGLE-BANK PENALTY · 2024

## $3.09B

Largest BSA enforcement action of 2024. Includes a U.S. asset cap (~$434B) — only the second ever on a major bank.

FinCEN · DOJ · OCC consent order Oct 2024

ANNUAL INDUSTRY SPEND · US + CA

## $61B

Annual cost of financial-crime compliance. The CCO's recurring board question — *"is the juice worth the squeeze?"* — is now FinCEN's literal RFI title.

LexisNexis Risk Solutions Feb 2024 · FinCEN Sep 2025 RFI

INDUSTRY FALSE-POSITIVE BASELINE

## 95%

Of every 100 alerts, 95 close as legitimate. Analyst attention is the scarce resource — and most of it is spent on noise, not signal.

Flagright industry analysis · NICE Actimize 2026

**KEY TAKEAWAY** The visible cost is the fine. The invisible cost is the analyst hour spent on the 95th false positive — every day, every quarter, every year.

**Source:** FinCEN consent-order release Oct 2024 · LexisNexis True Cost of FCC Feb 2024 · FinCEN AML-cost RFI Sep 2025 · Flagright TM industry analysis

# The diagnosis is in the language: four sentences a CCO says before the regulator does.

Quoted verbatim from public regulator letters and consent orders — none paraphrased. Every line below is something a Chief Compliance Officer or regulator has put on the record between 2024 and 2026.

PAIN-1 · AUDIT-DEFENSIBILITY

"Decisions made in relation to financial crime were not supported by evidence or an audit trail of debate and challenge."

**FCA** · Dear-CEO Letter to Annex 1 firms · Mar 2024

PAIN-2 · BACKLOG VISIBILITY

"Examiner testing only determined whether controls existed rather than if they were actually being used."

**K&L Gates** · 2024 BSA enforcement-action review

PAIN-6 · MODEL VALIDATION

"Transaction monitoring systems... meet the definition of 'model' and are held to the same standard."

**ACAMS / 2021 Interagency Statement** · still cited 2026

PAIN-8 · ROI

"Is the juice worth the squeeze?"

**FinCEN** · AML-Compliance-Cost RFI literal subject line · Sep 2025

**Source:** All quotes verbatim from cited regulator publications · full pain-point research at *docs/research/2026-04-aml-process-pain.md*

# One specification serves three reviewers — without re-translation.

One machine-readable file. Authored by Compliance, executed unchanged by Engineering, replayed byte-for-byte by an Auditor — every quarter.

SOURCE OF TRUTH

**Compliance Manifest**

written once · version-controlled · PR-reviewed

FIRST LINE · COMPLIANCE

The risk policy in machine-readable form, in our own language.

*"Our scenarios, our thresholds, every line cited to a regulation."*

ENGINEERING

The schema we build from. The data the engine consumes.

*"Compiles deterministically — same spec, same data, same answer."*

AUDITOR · 2LOD

The artifact I can replay any quarter, byte-for-byte.

*"When the regulator asks 'who decided?' — the answer is in the file."*

**Source:** AML Open Framework specification model · primary-source quotes from FCA Dear-CEO 2024 + K&L Gates 2024 BSA review



KEY TAKEAWAY Remove the translation layer between 1LoD / 2LoD / 3LoD — and you remove the most-cited finding of 2024–26.

# Five stages from one document to a regulator-ready audit.

One Manifest. Five deterministic transformations. Every artefact in the chain — alerts, cases, STRs, audit ledger entries — generated from the document, never hand-written in parallel.

Stage · 01

### Compliance Manifest

Compliance writes one document. Every rule cites the regulation that justifies it. PR-reviewed, version-controlled.

AUTHORED BY · CCO / MLRO

→

Stage · 02

### Generators

SQL queries, DAG stubs, data-quality tests, MRM dossiers, control matrix — all built from the Manifest.

OPERATED BY · 1LOD ENG

→

Stage · 03

### Engine

Rules execute on the warehouse. Alerts produced with the full evidence chain attached. ISO 20022 native.

OPERATED BY · 1LOD ENG

→

Stage · 04

### Cases & STRs

SLA-timed queues. Auto-drafted narratives. Regulator-ready ZIP via `aml audit-pack`.

OPERATED BY · 1LOD ANALYST

→

Stage · 05

### Audit ledger

Every decision SHA-256-hash-chained. Re-run any history byte-for-byte. Tampering breaks the chain.

VERIFIED BY · 2LOD / AUDIT

DETERMINISM GUARANTEE **same Manifest + same data + same seed = identical output hashes.** No commercial platform publishes this guarantee. SR 26-2 expects it from challenger models.

KEY TAKEAWAY The four other slides in this section show what each stage *removes* — analyst hours, cross-team meetings, audit-prep sprints. The pipeline above is what removes them.

**Source:** Reference architecture · 4-layer stack documented at *docs/architecture.md* · spec model in *spec/loader.py* · audit chain in *engine/audit.py* · re-runnability tested by *test_run_is_reproducible*

# Same Tuesday, two days. The unmeasured win is analyst hours.

Today | **Compare** | With framework

| | TODAY · WITHOUT THE FRAMEWORK | | SAME TUESDAY · WITH THE FRAMEWORK |
|---|---|---|---|
| 9 AM | ALERT TRIAGE<br>Open **8 systems** for one alert. Copy-paste evidence into Word. | 01 → | ALERT TRIAGE<br>One evidence bundle. Customer, transactions, prior alerts, network neighbours, regulator citation — all on one screen. |
| 11 AM | KYC REVIEW<br>Pull dossier from 3 systems. Justify risk in a free-text box. Annual-review backlog grows by month-end. | 02 → | KYC REVIEW<br>Risk-rating computed from the spec. Dossier auto-assembled. Reviews stop being a backlog. |
| 1 PM | STR DRAFTING<br>Hunt customer history + transaction trail + typology citation. Re-type the narrative in goAML. | 03 → | STR DRAFTING<br>Narrative auto-drafted from the alert evidence chain. Analyst reviews and signs. |
| 3 PM | BOI REFRESH<br>Email RM, chase a BO form, attach to the case. Stale list lives in someone's spreadsheet. | 04 → | BOI REFRESH<br>BOI Workflow page shows every reporting-company customer + freshness state. Stale BOI is an alert. |
| 5 PM | BACKLOG STATUS<br>"How many alerts are over SLA?" — *"I'll ask the team."* Dashboard shows last week's red. | 05 → | BACKLOG STATUS<br>SLA timer + escalation engine. Per-queue breach-rate badges. CCO sees real status first. |

**KEY TAKEAWAY** The unmeasured win isn't accuracy — it's *analyst hours*. Multiply across 100 analysts, 50 weeks: same dollars, signal instead of motion.

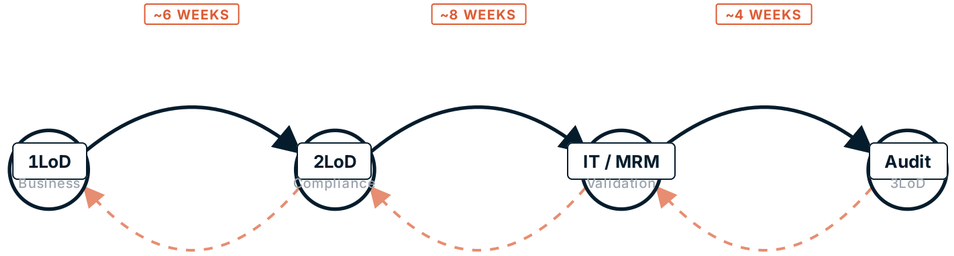
**Source:** Industry false-positive baseline · Flagright + NICE Actimize 2026 · STR-filing-latency p95 cited in 2024 BSA consent orders · framework workflows in *Investigations*, *BOI Workflow*, *Tuning Lab*, *Audit & Evidence* dashboard pages

# The other half of the win is political.

Today | **Compare** | With framework

TODAY · LINEAR HANDOFFS

**Months of meetings — until alignment, then ship.**




↻ ITERATE FOR MONTHS

until alignment, then ship

WITH FRAMEWORK · HUB & SPOKE

**Everyone reads the same artifact. *One PR.***

git diff · review · merge

audit chain records the change

| | | |
|---|---|---|
| Rule-change approval — months of 1LoD ⇄ 2LoD ⇄ IT alignment | → | One PR. Audit chain records the change. |
| BRA refresh — quarterly committee assembling spreadsheets | → | Live render from the Manifest. Framework Alignment page. |
| Annual MRM challenge — 4-week consultant engagement | → | Backtester self-serves. Per-rule precision/recall. |
| Pre-exam evidence — 8-person sprint, 6 weeks of forensics | → | `aml audit-pack` CLI exports the regulator-ready ZIP. |

**KEY TAKEAWAY** Detection vendors sell better alerts. The framework removes the *meetings about the alerts* — and that is where the program's calendar actually lives.

**Source:** K&L Gates 2024 BSA enforcement-action review · FCA Dear-CEO Letter Mar 2024 · framework workflows in *Spec Editor*, *Framework Alignment*, *Tuning Lab*, *Audit Evidence* dashboard pages · `aml audit-pack` CLI

# Each pain has a single, evidenced answer — one file away.

Direct mapping from public-record regulator findings to specific framework capabilities. Hover any card to trace the connection.

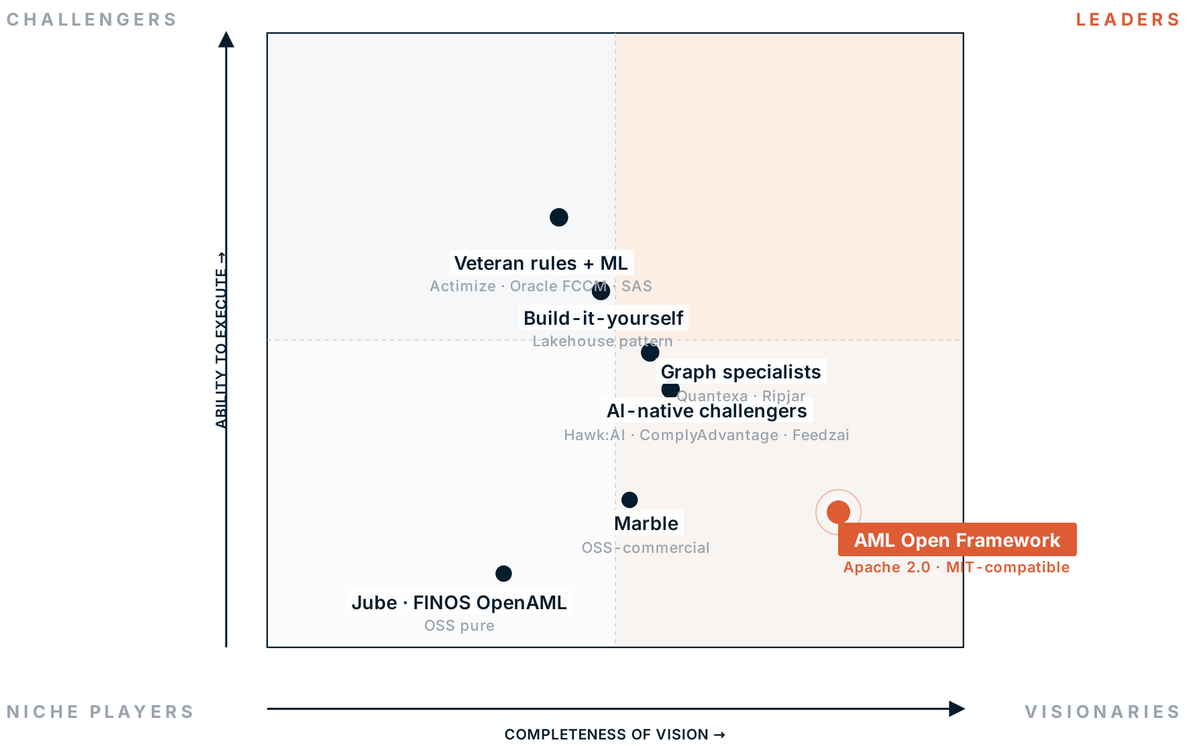
| LEADER'S PAIN · REGULATOR-CITED | | FRAMEWORK CAPABILITY · WHAT CHANGES |
|---|---|---|
| PAIN · 01<br>*"We can't prove what we did."*<br>FCA Dear-CEO · Mar 2024 | MAPS TO<br>One Manifest entry · one regulator citation · one audit-trail entry — for every pain. | CAPABILITY · 01<br>**Hash-chained audit ledger** + deterministic replay<br>Any historical run reproducible byte-for-byte. **Audit & Evidence** page. |
| PAIN · 02<br>*"The backlog has been red on the board for years."*<br>2024 BSA consent order | | CAPABILITY · 02<br>**SLA timer + escalation engine**<br>Red queues become loud, not slide-decoration. Per-queue breach-rate badges. **Investigations** page. |
| PAIN · 03<br>*"Our risk assessment is a PDF nobody trusts."*<br>FCA + FINTRAC 2025 AMP regime | | CAPABILITY · 03<br>The Manifest **IS** the risk assessment<br>Machine-readable, cited line-by-line. **Framework Alignment** page renders the live BRA. |
| PAIN · 04<br>*"My monitoring system is a model and I cannot validate it."*<br>ACAMS · 2021 Interagency Statement | | CAPABILITY · 04<br>**Per-rule MRM dossier** + 4-quarter backtester<br>"Is rule X still earning its keep?" answered before lunch — no vendor study. |

**KEY TAKEAWAY** Every pain on the left has a *specific Manifest entry + dashboard page + regulator citation* on the right. No black box, no vendor add-ons.

**Source:** Pain points 1, 2, 4, 6 from *docs/research/2026-04-aml-process-pain.md* · capability mapping in framework README + audit-evidence.md

# The deterministic + spec-as-source-of-truth corner is uncontested.

Plotted against six 2026 buyer archetypes on the two axes regulators care about — *completeness of vision* and *ability to execute*.




STRATEGIC PLANNING ASSUMPTION

By April 2026, SR 26-2 folds AML transaction monitoring into model-risk inventory — every **ML-first vendor carries a model-validation tax** on every typology.

- **AML Open Framework — Visionary, by design**
  Spec-as-policy is *not a model* under SR 26-2. Hash-chained replay; multi-jurisdiction templating; ISO 20022 in-tree. *Buyer that lands:* 2LoD MRM / Audit at any Cat-2+ bank.
- **Veteran platforms — Leaders by inertia**
  Win Tier-1 RFPs by default. Differentiator: scale + audit comfort. *Where we slot:* 2LoD challenger model for SR 26-2 independent re-implementation.
- **AI-native challengers — Visionary but ML-taxed**
  Hawk:AI Series C ($56M+$10M); ComplyAdvantage cites 65–85% FP reduction. *Adjacent, not opposed* — they win mid-market; we cover the same buyer only when the buyer wants source-available code.

KEY TAKEAWAY The framework doesn't compete with veteran platforms on detection scale — it competes on *provability*. Under SR 26-2, that is now the buyer's procurement criterion.

**Source:** Framework analysis based on cited 2026 vendor information (SymphonyAI, Alessa, Tookitaki, ComplyAdvantage, SiliconANGLE) · SR 26-2 letter Apr 2026 · OCC Bulletin 2026-13 · *Not a Gartner publication.*

# The framework arrives built, not blueprinted.

Open-source · runs in your perimeter · no per-seat licence · every rule cites a regulation · re-runs deterministically.

**8**

CURATED TYPOLOGIES

Every line cited to a regulator publication. Day-one detection.

**4**

ACTIVE JURISDICTIONS

US (FinCEN) · CA (FINTRAC) · UK (FCA) · EU (EBA). FATF Travel Rule.

**26**

DASHBOARD PAGES

Persona-aware. Sidebar filters by role — CCO sees board signal, analyst sees queue.

**9**

REFERENCE SPECS

Schedule-I bank · community bank · VASP · cyber-fraud · trade-based ML · …

**1,632**

TESTS · ALL GREEN

5 CI jobs per pull request. Lint, unit, API, e2e dashboard, docker.

**SHA-256**

AUDIT CHAIN

Hash-chained decision log. Tampering breaks attestation, by construction.

**MIT**

LICENCE

Apache 2.0 / MIT compatible. Forkable. Runs entirely in your perimeter.

**$0**

PER-SEAT FEES

No vendor licence renewals. Effectiveness pack quantifies what you spent.

**Source:** Live counts from *main* branch as of 2026-04-29 · github.com/tomqwu/aml_open_framework

THREE PATHS FROM HERE

# The next step takes five minutes.

WHERE YOU ARE
**Curious**

① READ · 15 MIN

## Show me the research.

Open the 10-pain-point doc with primary regulator citations + a 2024 enforcement case study. Unbranded frame, citable sources.

/RESEARCH/PROCESS-PAIN →

② TRY · 5 MIN

## Run it on my desk.

Clone, install, run a real audit pack against the canonical spec. `aml audit-pack` — no vendor demo cycle.

GITHUB.COM/TOMQWU/AML_OPEN_FRAMEWORK →

③ TALK · 30 MIN

## Walk me through it.

Bring your spec or pain — open a GitHub Discussion, pull-request a typology, or tag the maintainer in an issue.

GITHUB.COM/.../DISCUSSIONS →

OPEN SOURCE · MIT · RUNS IN YOUR PERIMETER

github.com/tomqwu/aml_open_framework

**Source:** AML Open Framework · primary-source research grounded in FCA / FinCEN / OSFI / FINTRAC public record